CLASS 6 — MATHS · HARD LEVEL FULL-YEAR PAPER SET 1

Full Syllabus (Chapters 1–10) | RBSE Pattern | Session 2026–27

Time: 3 Hours Name: _____ Roll No.: _____ **Maximum Marks: 80**

General Instructions:

1. All questions are compulsory.
2. Q. 1–20 are Objective Type (1 mark each).
3. Q. 21–26 are Very Short Answer Type (2 marks each).
4. Q. 27–32 are Short Answer Type (3 marks each).
5. Q. 33–38 are Long Answer Type (5 marks each).
6. Internal choice is given in Q. 31–32 and Q. 36–38.
7. Write the question number before each answer.
8. Hard level: application and higher-order questions.

SECTION — A (Objective Type Questions) 20 × 1 = 20

Choose the correct option and write it in your answer book.

1. At a party, 15 people each hug every other person exactly once. Number of hugs: 1

(a) 112 **(b)** 120 **(c)** 105 **(d)** 210

2. A clock shows 2 o'clock. After 3 hours, the smaller angle between the clock hands is: 1

(a) 150 **(b)** 210 **(c)** 60 **(d)** 90

3. How many steps does the Kaprekar routine take to reach 6174 from 3141? 1

(a) 6 **(b)** 5 **(c)** 7 **(d)** 4

4. The total of the table, after removing the largest value, is: 1

Item	Number
Banana	75
Mango	25
Orange	15
Apple	55
Grapes	45

(a) 200 **(b)** 140 **(c)** 145 **(d)** 215

5. The smallest 4-digit number divisible by 13 is: 1

(a) 988 **(b)** 1002 **(c)** 1014 **(d)** 1001

6. A rectangle's length is 2 times its breadth and its perimeter is 54 cm. Its area (cm^2) is: 1

(a) 180 **(b)** 153 **(c)** 171 **(d)** 162

7. $\frac{5}{6} + \frac{3}{8} - \frac{1}{4} =$ 1

(a) $\frac{23}{24}$ **(b)** $1\frac{11}{24}$ **(c)** $\frac{17}{24}$ **(d)** $1\frac{1}{6}$

8. Two concentric circles have radii 4 cm and 9 cm. A point is 0 cm from the common centre. It lies in the ring between the circles only if its distance is more than 4 cm and less than 9 cm. How many of the distances 3, 4, 6, 9, 8, 12 cm give a point in the ring? 1

(a) 4 **(b)** 2 **(c)** 6 **(d)** 3

9. A regular polygon has an angle of rotation 40° and its perimeter is 45 cm. The side length is (cm): 1

(a) 7 **(b)** 6 **(c)** 5 **(d)** 4

10. -14 - (-9) + (22) - (-5) = 1

(a) 32 **(b)** 24 **(c)** 20 **(d)** 22

11. If $1 + 2 + 3 + \cdots + n = 703$, then n is: 1

(a) 36 **(b)** 19 **(c)** 37 **(d)** 38

12. A wheel turns 3 full turns and then a further quarter turn. The total angle in degrees is: 1

(a) 1170 **(b)** 1080 **(c)** 1350 **(d)** 1260

13. The hailstone (Collatz) sequence of 25 has how many terms (including 25 and 1)? 1

(a) 23 **(b)** 22 **(c)** 25 **(d)** 24

14. One symbol of a pictograph stands for 10. Class A has 12 symbols and its total is 3 times Class B. How many symbols does Class B have? 1

(a) 5 **(b)** 9 **(c)** 36 **(d)** 4

15. The number of composite numbers from 1 to 30 (both included) is: 1

(a) 18 **(b)** 19 **(c)** 21 **(d)** 20

16. A square and an equilateral triangle have equal perimeters. If the square's side is 12 cm, the triangle's side (cm) is: 1

(a) 12 **(b)** 16 **(c)** 28 **(d)** 20

17. **Assertion (A):** In the sequence $3, 7, 11, 15, \ldots$ the next term is 19. 1
Reason (R): Each term of this sequence is obtained by adding 4 to the term before it.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

18. **Assertion (A):** An angle of 135° is an obtuse angle. 1
Reason (R): An obtuse angle is more than 90° and less than 180°.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

19. **Assertion (A):** The largest number that can be made from the digits 3, 0, 7, 5 (each used once) is 7530. 1
Reason (R): To make the largest number, arrange the digits in descending order.
Choose the correct option:

(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

20. **Assertion (A):** In a pictograph where one symbol stands for 10 children, 3 and a half symbols show 35 children. 1
Reason (R): A half symbol stands for half of the value of a whole symbol.
Choose the correct option:
(a) Both A and R are true and R is the correct explanation of A.
(b) Both A and R are true but R is not the correct explanation of A.
(c) A is true but R is false.
(d) A is false but R is true.

SECTION — B (Very Short Answer Type) 6 × 2 = 12

Answer in about 20–30 words each.

21. 14 cricket teams play a league in which every team plays every other team once. Find the number of matches. 2

22. Find the smaller angle between the clock hands at 4 o'clock, and the angle the hour hand turns through from 4 o'clock to 6 o'clock. 2

23. How many steps does the Kaprekar routine need for 1234? Write the chain. 2

24. In a survey of 150 people, 62 like tea, 48 like coffee, and the rest like milk. Find the number liking milk and show data in a table. 2

25. Find the smallest number divisible by 8, 12 and 18. 2

26. The length and breadth of a rectangle are in the ratio 3:2 and its perimeter is 60 cm. Find the sides. 2

SECTION — C (Short Answer Type) 6 × 3 = 18

Answer in about 40–60 words each.

27. In a pattern the 1st term is 14, the 2nd is 23, and so on by adding 9. Which term of the pattern equals 365? Find also the term just before it. 3

28. At 1 o'clock the hands of a clock make a smaller and a larger angle. Find both and say which one is a reflex angle (if any). 3

29. Find a 4-digit palindrome of the form $abba$ with $a = 1$ and $b = 5$. Show that it is divisible by 11. 3

30. The table shows two weeks of sales (units). Find the week with the larger total and by how much. 3

Day	Week 1	Week 2
Mon	13	15
Tue	20	19
Wed	11	14
Thu	17	12

31. Three traffic lights change every 48, 72 and 108 seconds. If they change together at 8:00 am, after how many seconds do they next change together and at what time? 3

OR

A rectangular plot 40 m by 25 m is to be fenced with 3 rounds of barbed wire costing ₹12 per metre. Find the total cost. Also find the cost of ploughing it at ₹4 per m^2. 3

32. A school has 480 students. $\frac{5}{8}$ of them are girls. $\frac{2}{5}$ of the girls play chess. How many girls play chess and how many boys are there? 3

OR

Four circles of radius 3 cm are placed at the corners of a square so that adjacent circles touch. Find the side of the square joining centres and its area. 3

SECTION — D (Long Answer Type) 6 × 5 = 30

Answer in about 100–150 words each.

33. **Case study.** Ravi makes a row of squares with matchsticks. One square needs 4 sticks, two squares joined side by side need 7 sticks, and three squares need 10 sticks. He keeps adding one square at a time. 5

(i) How many sticks are needed for 4 squares?

(ii) How many sticks does each new square add?

(iii) How many sticks are needed for 10 squares?

(iv) Ravi has exactly 40 sticks. How many squares can he make in a row? Show how you found it.

34. **Case study.** The clock on the wall of Anil's school shows 2 o'clock. The hour hand turns slowly, and the minute hand moves faster. 5

(i) Through how many degrees does the minute hand turn in 20 minutes?

(ii) Through how many degrees does the hour hand turn in 1 hour?

(iii) What is the type of the angle between the hands at 4 o'clock?

(iv) Find the smaller angle between the hands at 7 o'clock and name the type of the other angle formed.

35. A palindromic number of 5 digits is formed as $abcba$ where $a = 1$, $b = 2$, $c = 3$. Write the number, find its digit sum, check if it is divisible by 9, and find the smallest number to add to make it divisible by 9. 5

36. A bar graph has scale 1 cm = 12 units. Bars A, B, C have heights 3 cm, 6 cm, 9 cm. The value of D equals A + B - C. Find the values of A, B, C and D, and the height of bar D. 5

OR

A rectangular garden is 60 m by 36 m. Square flower beds of the largest possible equal size are to be made covering the garden. Find 5

(i) the side of each bed,

(ii) the number of beds,

(iii) the cost of fencing all beds at ₹5 per metre if each bed is fenced separately.

37. A rectangle with length 20 cm and breadth 12 cm has a triangular piece with base 20 cm and height 8 cm cut from one side. Find the area of the remaining piece and the fraction of the whole that is left. 5

OR

A trader sold $\frac{1}{3}$ of his cloth on day 1 and $\frac{1}{4}$ of the remaining on day 2. If 30 metres are left, find the total length of cloth. 5

38. Describe the construction of a rectangle with diagonal 10 cm and one side 6 cm (you may use a compass for arcs). Find the other side if it is 8 cm, check using the property that the half-diagonal is 5 cm, and find the perimeter. 5

OR

For a regular 6-gon of side 5 cm: 5

(a) find the number of lines of symmetry,

(b) the angle of rotation,

(c) the perimeter,

(d) the angle turned by a vertex when the polygon is rotated 4 times through the smallest angle.

END OF QUESTION PAPER **Total: 80 Marks**

MASTERMIND